FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
CONCEPT RECAPITULATION TEST – I
PAPER –1
TEST DATE: 01-05-2024

ANSWERS, HINTS & SOLUTIONS

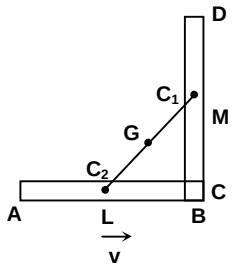
Physics

PART – I

SECTION – A

1. B, C

Sol. Conservation of angular momentum about G gives:



$$Mv\cdot\frac{L}{4}=2\times\left[\frac{1}{12}ML^2+M\left(\frac{\sqrt{2}L}{4}\right)^2\right]\times\omega$$

Solving

$$\omega=\frac{3v}{5L}$$

conservation of momentum gives

$Mv = 2Mv_{CM}$

or, $v_{CM} = v/2$

2. A, C

Sol. $\frac{dv}{dx}=\frac{2}{1}=2s^{-1}$

$$F=\eta A\frac{dv}{dx}=10^{-3}\times10\times2=0.02\text{ N}$$

3. A, B, C

Sol. The tangential acceleration is $-\mu g$

$$v_0-\mu gt=0$$

$$t=\frac{v_0}{\mu g}$$

Centripetal acceleration = $\frac{k}{m}v$

$$\tan 60 = \frac{a_c}{a_t} = \frac{k}{m}\frac{v_1}{\mu g}$$

$$v_1 = \frac{\mu mg\sqrt{3}}{k}$$

$$\tan 30 = \frac{kv_2}{m\mu g}$$

$$v_2 = \frac{\mu mg}{\sqrt{3}k}$$

$$\frac{v_1 - v_2}{\mu g} = t = \frac{m}{k}\frac{2}{\sqrt{3}}$$

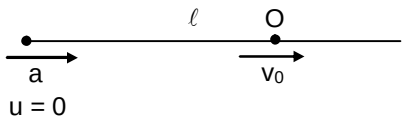
$$\frac{1}{2}mv_0^2 = \mu mgs$$

$$s = \frac{v_0^{\ 2}}{2\mu g}$$

$$R = \frac{v^2}{a_c} = \frac{m}{k}(v_0 - \mu gt)$$

4. B

Sol. Let first pulse be released at t = 0.



Time when first pulse reaches O = $t_1 = \frac{\ell}{v - v_o}$

Time when second pulse reaches O =

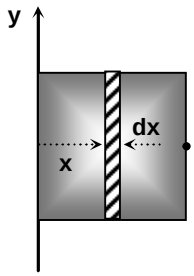
$$t_2 = T + \frac{\ell + v_o T - \frac{1}{2}aT^2}{v - v_o}$$

$$T' = t_2 - t_1 = \frac{vT}{v - v_o} - \frac{aT^2}{2(v - v_o)}$$

$$\therefore \quad f' = \frac{2f^2(v - v_o)}{2fv - a}$$

5. C

Sol. $dM = \sigma_0\left(1 - \frac{x}{a}\right)dx$



$$M = \frac{\sigma_0 a^2}{2}$$

$$\Rightarrow d(MOI) = dmx^2$$

$$MOI = \int dMx^2 = \frac{Ma^2}{6}$$

6. A

Sol. $T = kp^x d^y E^z$

$$[T] = [ML^{-1}T^{-2}]^x[ML^{-3}]^y[ML^2T^{-2}]^z$$

$x + y + z = 0$ ...(1)

$-x - 3y + 2z = 0$ ...(2)

$-2x - 2z = 1$ ...(3)

$x + z = -\frac{1}{2}$ ...(4)

$-y = -\frac{1}{2}$

$\Rightarrow \quad y = \frac{1}{2}$

by equation (2) $-x - \frac{3}{2} + 2z = 0$

$-x + 2z = \frac{3}{2}$ ...(5)

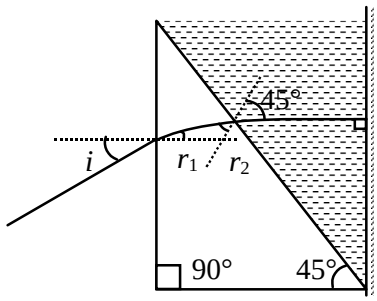
Adding (4) & (5)

$3z = 1,\ z = \frac{1}{3}$

$\Rightarrow \quad x = -\frac{1}{2} - \frac{1}{3} = -\frac{5}{6}$

7. B

Sol. Since incident ray retraces its path it must strike the plane mirror perpendicularly.



From Snell's law $\sin i = \mu_1 \sin r_1$

and $\mu_1 \sin r_2 = \mu_2 \sin 45° \Rightarrow \mu_1 \sin r_2 = \frac{\mu_2}{\sqrt{2}}$

$\Rightarrow \quad r_2 = \sin^{-1}\left(\frac{\mu_2}{\sqrt{2}\mu_1}\right)$

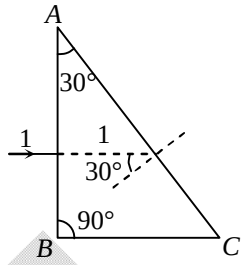
Also, $r_1 + r_2 = \frac{\pi}{4}$

$\therefore \quad r_1 = \frac{\pi}{4} - \sin^{-1}\left(\frac{\mu_2}{\sqrt{2}\mu_1}\right)$

$\therefore \quad i = \sin^{-1}\left[\mu_1 \sin\left(\frac{\pi}{4} - \sin^{-1}\frac{\mu_2}{\sqrt{2}\mu_1}\right)\right]$

8. B

Sol. (P) Refractive index of the prism is the minimum value required for ray (1) to undergo total internal reflection at face *AC*. Ray (1) falls on face *AC* at an angle of incidence 30°



$\therefore$ $30° > i_C$

$\sin 30° > \sin i_C$

$\therefore$ $\mu > 2$

Minimum value of μ can be taken as 2.

(Q) For ray 2, refractive angle of prism is 30°. Apply Snell's law for refraction at face *AB*.

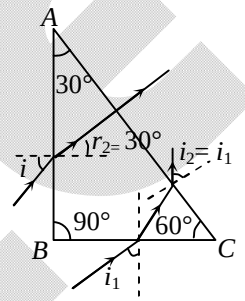
$1 \sin i = \mu \sin r$

$i = 90°$

(R) Using the relation $i_1 + i_2 = A + \delta$ for ray 2.

$90° + 0° = 30° + \delta$

$\delta = 60°$



(S) $\mu = \dfrac{\sin\left(\dfrac{A+\delta m}{2}\right)}{\sin\dfrac{A}{2}} \Rightarrow \delta m = 120°$

9. D

Sol. $w_g = \vec{F}.\vec{S} = (mg\sin\theta)S = 2\times 10\times\frac{1}{2}\times\frac{20}{100} = 2\,\text{J}$

$w_s = -\frac{1}{2}kx^2 = -\frac{1}{2}(1000)\left(\frac{20}{100}\right)^2 = -20\,\text{J}$

$w_N = 0$

From work energy theorem $w_g + w_{sp} + w_N + w_{ex} = \Delta k$ or $2 - 20 + 0 + w_{ex} = 0$

$w_{ex} = 18$ J

10. A

Sol. From *A* to *B*

Angular frequency $\omega = \sqrt{K/M}$

$V = \omega\sqrt{A_1^2 - X^2} \quad \Rightarrow \quad V^2 = \dfrac{K}{M}\left(A_1^2 - L^2\right)$

$\Rightarrow \quad A_1 = \left[\dfrac{MV^2}{K} + L^2\right]^{1/2} = \sqrt{2} \quad \ldots(\text{i})$

Also, $L = A_1\cos\omega t$, $\quad t_1 = \sqrt{\dfrac{M}{K}}\cos^{-1}\left[\dfrac{L}{\left(L^2 + \dfrac{MV^2}{K}\right)^{1/2}}\right] = \pi/4$

From *B* to *O*

Now angular frequency $\omega_2 = \sqrt{\dfrac{K}{2M}}$, $\quad V = \omega_2\left(A_2^2 - L^2\right)^{1/2}$,

$$A_2 = \left[L^2 + \frac{2MV^2}{K} \right] = \sqrt{3}$$

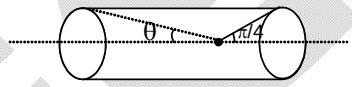
Also $L = A_2 \sin \omega t_2$, $\quad t_2 = \sqrt{\frac{2M}{K}} \sin^{-1} \frac{L}{\left[L^2 + \frac{2MV^2}{K} \right]^{1/2}} = \sqrt{2} \sin^{-1} \frac{1}{\sqrt{3}}$

Now V_{cm} at O is $A_2\omega_2 = \sqrt{\frac{3}{2}}$

The maximum compression of the spring is $\sqrt{3}$

11. D

Sol. $\tan\theta = \frac{R}{2R} = \frac{1}{2}$



$$\cos\theta = \frac{2}{\sqrt{5}}$$

Flux through F_1, $\phi_1 = \frac{q}{2\varepsilon_0}\left[1 - \cos\frac{\pi}{4}\right] = \frac{q}{2\varepsilon_0}\left[1 - \frac{1}{\sqrt{2}}\right]$

Flux through F_2, $\phi_2 = \frac{q}{2\varepsilon_0}\left[1 - \cos\theta\right] = \frac{q}{2\varepsilon_0}\left[1 - \frac{2}{\sqrt{5}}\right]$

Flux through cylinder $\phi = q/\varepsilon_0$

Flux through curved surface $= \phi - \phi_1 - \phi_2$

$$= \frac{q}{\varepsilon_0}\left[1 - \frac{1}{2} + \frac{1}{2\sqrt{2}} - \frac{1}{2} + \frac{1}{\sqrt{5}}\right] = \frac{q}{\varepsilon_0}\left[\frac{1}{2\sqrt{2}} + \frac{1}{\sqrt{5}}\right]$$

SECTION – B

12. 1

Sol. $P = \frac{1}{2}\rho\omega^2 A^2 sV$

Since $\frac{\lambda_1}{\lambda_2} = \frac{1}{2}, \frac{f_1}{f_2} = \frac{\omega_1}{\omega_2} = \frac{2}{1}$

Since $P_1 = P_2, \omega_1 A_1 = \omega_2 A_2$,

$$\frac{A_1}{A_2} = \frac{\omega_2}{\omega_1} = \frac{1}{2}$$

Pressure amplitude, $P_0 = B_0 Ak$

$$(P_0)_1 / (P_0)_2 = \left(\frac{A_1}{A_2}\right)\left(\frac{k_1}{k_2}\right) = \left(\frac{A_1}{A_2}\right)\left(\frac{\lambda_2}{\lambda_1}\right) = \left(\frac{1}{2}\right)\left(\frac{2}{1}\right) = 1$$

13. 100

Sol. $B = -\dfrac{P}{\Delta V / V}$ or $P = -B.\dfrac{\Delta V}{V}$

$\therefore \quad P = 9.8\times10^8 \times \left(\dfrac{0.1}{100}\right)$

or $\quad h\rho g = 9.8\times10^8 \times \left(\dfrac{0.1}{100}\right)$

$\therefore \quad h = \dfrac{9.8\times10^8}{1000\times9.8}\times\dfrac{0.1}{100}$ m = 100 m

14. 6

Sol. As source (horn of bus) is approaching stationary wall (say, listener), therefore, apparent frequency striking the wall is

$$v' = \frac{\upsilon\, v}{\upsilon - \upsilon_s} \qquad \ldots(1)$$

Sound of this frequency will be reflected by the wall (now, source). The passenger is the listener moving towards source. Therefore, frequency heard by the listener $v'' = \dfrac{(\upsilon + \upsilon_L)v'}{\upsilon}$

Using (1)

$$v'' = \frac{\upsilon + \upsilon_L}{\upsilon}\times\frac{\upsilon v}{\upsilon - \upsilon_s} = \frac{(\upsilon + \upsilon_L)v}{\upsilon - \upsilon_s}$$

$$= \frac{(330+5)\times200}{330-5} = \frac{335}{325}\times200$$

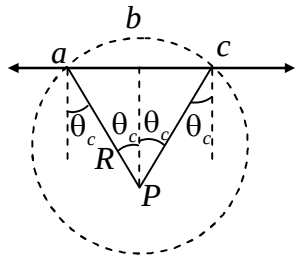
$v'' = 206\,\text{Hz}$

$\therefore$ Beat frequency = $(v''-v)$

$= 206 - 200 = 6\,\text{Hz}$

15. 1

Sol. The light escape is confined within a cone of apex angle '$2\theta_c$' where θ_c is the critical angle. Imagine a sphere with source of light as its centre and the surface area *abc* is *A*.



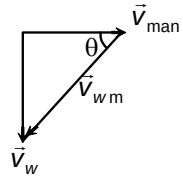
here $A = \int_0^{\theta_c} 2\pi R^2 \sin\theta\, d\theta = 2\pi R^2(1-\cos\theta_c)$

$= \pi R^2 \qquad \left[\therefore \theta_c = \sin^{-1}\left(\dfrac{\sqrt{3}}{2}\right) = 60^o\right]$

$\therefore$ Power transfer $= P\times\dfrac{A}{4\pi R^2} = = 4\times\dfrac{1}{4} = 1\,\text{W}$

16. 100

Sol. Let velocity of wind $= v_w$



Initially velocity of man = 2.5 π m/s

Velocity of wind with respect to man makes an angle 45^0 with track initially

$$\tan\theta = \frac{v_w}{v_{man}}$$

$\theta = 45^0$ when $v_{man} = 2.5\,\pi$ m/s $\Rightarrow$ $v_w = 2.5\ \pi$ m/s

Let when velocity of man is v_m the wind appears to make an angle θ with track, so

$$v_m = v_w \cot\theta$$

$$v_m = 2.5\pi\cot\theta$$

$$\frac{dx}{dt} = 2.5\pi\cot\theta \quad \text{[where } x \text{ is position of man w.r.t. origin at time } t\text{]}$$

$$\Rightarrow \quad \frac{dx}{d\theta}.\frac{d\theta}{dt} = 2.5\pi\cot\theta$$

but $$\frac{d\theta}{dt} = \frac{\Delta\theta}{\Delta t} = \frac{\pi}{4\left(\frac{10}{\ln\sqrt{2}}\right)} = \frac{\pi\ln\sqrt{2}}{40}$$

$$\frac{dx}{d\theta} = \frac{100\pi}{\pi\ln\sqrt{2}}\cot\theta$$

$$x = \frac{100\pi}{\pi\ln\sqrt{2}}\left[\ln\sin\theta\right]_{\pi/4}^{\pi/2} = \frac{100\pi}{\pi\ln\sqrt{2}}\left[\ln\sqrt{2}\right] = 100 \text{ m}$$

So length of race is 100 m.

17. 350

Sol. Time period of oscillation of R

$$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\times\frac{10}{\pi} = 20s$$

At a time t = 10 s, R will be at mean position and moving along negative x-axis.

$$v_R = A\omega = 15 \text{ m/s}$$

The sound which is received at t = 10s, is emitted at $t = t_0$ s.

$$\frac{1}{2}at_0^2 = v(10 - t_0)$$

$$\frac{1}{2}\times 18.75t_0^2 = 300(10 - t_0) \Rightarrow t_0 = 8\text{s}$$

$$v_s = at = 18.75\times 8 = 150 \text{ m/s}$$

$$f' = 500\left(\frac{300+15}{300+150}\right) = 350\text{Hz}$$

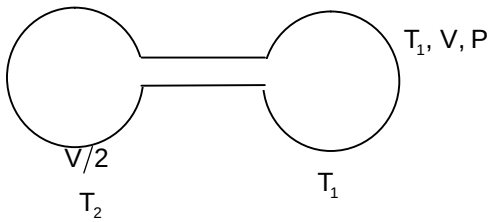
Chemistry

PART – II

SECTION – A

18. A,B,C,D
Sol. factual

19. A
Sol.



$PV = nRT$

$n_1 = \frac{PV}{RT_1}$

$n_2 = \frac{PV/2}{RT_2}$

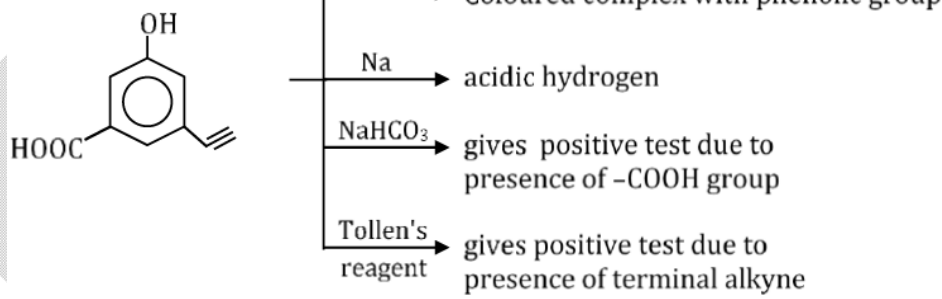
$P_1\left(\frac{V}{RT_1} + \frac{V}{2RT_2}\right) = \frac{P(V)}{RT_1}$

$P_1\left(1 + \frac{T_1}{2T_2}\right) = P$

$P_1\left(\frac{2T_2 + T_1}{2T_2}\right) = P$

$P_1 = \frac{2PT_2}{T_1 + 2T_2}$

20. A, B, C, D
Sol.



21. D
Sol. Using MOT of O_2 molecule.

22. C
Sol. At 2nd equation point:

$P^H = \frac{1}{2}\left(PK_{a_2} + PK_{a_3}\right) = \frac{8+12}{12} = 10$

Now, $K_{a_1}.K_{a_2}.K_{a_3} = \frac{\left[H^+\right]\left[A^{3-}\right]}{\left[H_3A\right]}$

or $7.5\times10^{-4}\times10^{-8}\times10^{-12} = \frac{\left(10^{-10}\right)\times\left[A^{3-}\right]}{\left[H_3A\right]}$

$\therefore \frac{\left[H_3A\right]}{\left[A^{3-}\right]} = \frac{10^{-6}}{7.5} = 1.33\times10^{-7}$

23. B

Sol. The mole fraction of A in distillate,

$$X_A^{'} = Y_A = \frac{X_A.P_A^{\circ}}{P_{total}} = \frac{\frac{1}{4}\times100}{\frac{1}{4}\times100+\frac{3}{4}\times80} = \frac{5}{17}$$

Now, V.P. of distillate, $P = X_A^{'}.P_A^{\circ} + X_B^{'}.P_B^{\circ}$

$= \frac{5}{17}\times100+\frac{12}{17}\times80 = 85.88\,\text{mm Kg}$

24. B

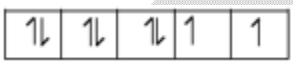
Sol. $Cu^{+2} + 2CN^{-1} \rightarrow CuCN + \frac{1}{2}(CN)_2\,\text{gas}$

25. A

Sol. Cr_2O_3- Amphoteric oxide
N_2O- Neutral oxide
Fe_3O_4- Mix oxide
CrO_3- Acidic oxide

26. C

Sol. $Ni^{2+} \rightarrow 3d^8\,4s^0$



$n \rightarrow 2$

$\mu = \sqrt{n(n+2)}$

$Cr^{2+} \rightarrow 3d^4 \rightarrow d-d$ transition because in aq. medium it exists as $[Cr(H_2O)_6]^{2+}$

$V^{2+} \rightarrow d^3 \rightarrow t_{2g}^3 e_g^0$

$Ti^{4+} \rightarrow [Ar]3d^0\,4s^0 \rightarrow$ no unpaired electron

27. A

Sol. (1) $[Cr(H_2O)_4Br_2]^+ \Rightarrow$ Paramagnetic, d^2sp^3, show geometrical isomerism
(2) $[Cu(NH_2CH_2CH_2NH_2]\,(CN)_2Cl]^{2-} \Rightarrow$ Paramagnetic, sp^3d^2, show Geometrical isomerism
(3) $[Pt(ox)_2]^{2-} \Rightarrow$ Diamagnetic, dsp^2
(4) $[Fe(OH)_4]^- \Rightarrow$ Paramagnetic, sp^3

28. A

Sol. HCO_3^-, CO_3^{2-} - CO_2 type gas with dil. H_2SO_4

Both Can't react with $K_2Cr_2O_7$ solution (acidic) due to + 4 oxidation of C in HCO_3^-, CO_3^{2-}

SECTION – B

29. 2

Sol. $n = 1 \Rightarrow t_{100\%} = \frac{1}{K}.\ln\frac{[A_0]}{0} = \text{Infinite}$

$n \neq t_{100\%} = \frac{[A_0]^{1-n} - (0)^{1-n}}{K(1-n)} = \frac{[A_0]^{1-n}}{K(1-n)}$ if $n < 1$

Infinite if $n > 1$

30. 5

Sol. $t_{1/2} = \frac{0.693}{6.93 \times 10^{-4}} = 1000 \text{ sec}$

	A	⟶	nB
$t = 0$	a mole		0
$t = 1000$ sec	$\frac{a}{2}$ mole		$\frac{n.a}{2}$ mole

Now, $\frac{\frac{a}{2} + \frac{n.a}{2}}{a} = 3 \Rightarrow n = 5$

31. 1520

Sol. $SnCl_2 \longrightarrow Sn^{2+} + 2Cl^-$

Cathode: $Sn^{2+} + 2e^- \longrightarrow Sn$

Anode: $2Cl^- \longrightarrow Cl_2 + 2e-$

$Cl_2 + SnCl_2 \longrightarrow SnCl_4$

Moles of $SnCl_2$ taken = $\frac{19}{190} = 0.1$

Moles of Sn produce = $\frac{1.19}{119} = 0.01$

= moles of Cl_2 produced

= moles of $SnCl_4$ formed

and moles of $SnCl_4$ left = 0.1 – (0.01 + 0.01) = 0.08

$\therefore \frac{m_{SnCl_2}}{m_{SnCl_4}} = \frac{0.08 \times 190}{0.01 \times 261} = \frac{1520}{261}$

32. 108

Sol. $\Lambda^\circ_{eq}\left[Ba_3(PO_4)_2\right] = 160 + 140 - 100$

= 200 Ohm^{-1} cm^2 eq^{-1}

Now, $\wedge^\circ_{eq} = \wedge_{eq} = \frac{\kappa}{C} \Rightarrow 200 = \frac{1.2 \times 10^{-5}}{S}$

$\therefore S = 6 \times 10^{-8}$ eq / cm^3 $= 6 \times 10^{-5}$N $= 10^{-5}$M

$\therefore K_{sp} = 108\ S^5 = 108 \times 10^{-25} M^5$

33. 5

Sol. Using MOT

2,1,2 Pia (π) bond in given C_2, O_2, N_2

34. 5

Sol. Carbon already exit in higher oxidation state (+ 4)

Mathematics

PART – III

SECTION – A

35. A, C, D

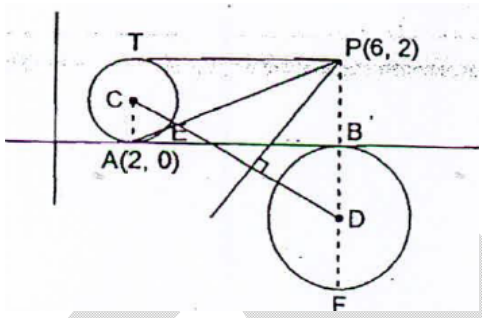
Sol. Clearly P lies on radical axis

$\Rightarrow PE \cdot PA = PB \cdot PF$

$\Rightarrow \frac{PE}{PF} = \frac{PB}{PA} = \frac{2}{2\sqrt{5}} = \frac{1}{\sqrt{5}}$

Again $PE \cdot PA = PT^2$

$\Rightarrow PE = \frac{4^2}{2\sqrt{5}} = \frac{8}{\sqrt{5}}$



36. A, B, D

Sol. Given $a \neq b \neq c$, $a, b, c \in R$

Now $ax^2 + bx + c \geq 0 \Rightarrow b^2 - 4ac \leq 0$ and $a > 0$ ....(1)

$bx^2 + cx + a \geq 0 \Rightarrow c^2 - 4ab \leq 0$ and $b > 0$ ....(2)

$cx^2 + ax + b \leq 0 \Rightarrow a^2 - 4bc \leq 0$ and $c > 0$ ....(3)

Equality cannot hold simultaneously ($\because a \neq b \neq c$)

$\therefore \frac{a^2 + b^2 + c^2}{ab + bc + ca} < 4$

and since $a \neq b \neq c$

$\Rightarrow (a-b)^2 + (b-c)^2 + (c-a)^2 > 0$

$\Rightarrow \frac{a^2 + b^2 + c^2}{ab + bc + ca} > 1$

$\Rightarrow \frac{a^2 + b^2 + c^2}{ab + bc + ca} \in (1, 4)$

37. A, B

Sol. Required probability = $\frac{{}^5C_4 (2)^4}{{}^{10}C_4} + \frac{\left({}^5C_1 \times 1\right) \times \left({}^4C_2 \times 2^2\right)}{{}^{10}C_4} \times \frac{\left({}^2C_2 \times 2^2\right)}{{}^6C_4} = \frac{8}{15}$

38. B

Sol. The locus will be circle with diameter with (1, 2) and (2, 3)

39. C

Sol. $V = \frac{1}{6} abd \sin\theta$

40. D

Sol. $f'(x) = -(2\cos(\sin x)\sin\sin(x)\cos x + 2\sin(\cos)\cos(\cos x)\sin x) \leq 0$

$\Rightarrow$ Range is $\left[f\left(\frac{\pi}{2}\right), f(0) \right]$ as f(x) is periodic with period $\frac{\pi}{2}$

41. D

Sol. As $(a + b + c)^3 = a^3 + b^3 + c^3 \Rightarrow (a + b)(b + c)(c + a) = 0$

$\Rightarrow (\sin x + \sin 2x)(\sin 2x + \sin 3x)(\sin 3x + \sin x) = 0$

$\Rightarrow 2\sin\frac{3x}{2}\cos\frac{x}{2}$ or $2\sin\frac{5x}{2}\cos\frac{x}{2} = 0$ or $2\sin 2x\cos x = 0$

$\Rightarrow$ On solving least positive angle is $\frac{2\pi}{5}$.

42. D

Sol. $P \to 2;\ Q \to 1;\ R \to 4;\ S \to 3$

43. D

Sol. (P) no. of digits of form $a_1 > a_2 > a_3 > a_4 = {}^{10}C_4 \times 1$

no. of digits of form $a_1 > a_2 = a_3 > a_4 = {}^{10}C_3 \times 1$

no. of digits = ${}^{10}C_4 + {}^{10}C_3 = 330$

(Q) $4\lambda+2 = 2(2\lambda+1) = 2 \times$ odd no

$\Rightarrow$ no. of division = $(8+1)\times(6+1)-1 = 62$

(R) Let E_i is the set which contains all possible function in which $x_i = y_i$ from inclusion exclusion

$n(UE_i) = \sum(nE_i) - \sum n(E_i \cap E_j) + \sum n(E_i \cap E_j \cap E_k) - .....$

$n(E_i) = 5^9 \quad n(E_i \cap E_j) = 5^8 \quad n(E_i \cap E_j \cap E_k) = 5^7$. & so on

$n(UE_i) = {}^5C_1 5^9 - {}^5C_2 5^8 + {}^5C_3 5^7 - {}^5C_4 5^6 + {}^5C_5 5^5$

no. of desired fn. = 510-n (UE^i).

(S) Let the sides of polygon are n.

So no. of diagonals ${}^nC_2 - n = 35$

$\Rightarrow$ n= 10

$\Rightarrow$ no. of triangles can be formed is = ${}^{10}C_3 = 120$

44. D

Sol. (P) $\alpha + \beta = \omega + \omega^2 + \omega^3 + \omega^4 + \omega^5 + \omega^6 = 0$

(Q) $\alpha\beta = (\omega + \omega^2 + \omega^4)(\omega^3 + \omega^5 + \omega^6) = 3\omega$

$(\alpha\beta)^3 = 27\omega^3 = 27$

(R) $\alpha = -1 + \omega = -1 + \frac{(-1 \pm \sqrt{3}\,i)}{2}$

(S) $\sum_{k=0}^{6} \omega^{k^2} = 1 + \omega^2 + \omega^4 + \omega^9 + \omega^{16} + \omega^{25} + \omega^{36}$

$= 3 + 4\omega = 3 + 4\frac{(-1 \pm \sqrt{3}\,i)}{2} = 1 \pm 2\sqrt{3}\,i = 1 \pm \sqrt{8}\,i$

45. B

Sol. (P) $(t_1^2, 2t_1), Q(t_2^2, 2t_2)$ and $t_2 = -t_1 - \frac{2}{t_1}$

$\tan\alpha = \frac{2}{t_1 + t_2} = \frac{2}{-\frac{2}{t_1}} = -t_1$

$\tan\beta = -t_2 = t_1 + \frac{2}{t_1}$

(Q) $\left(\frac{x+y+1}{\sqrt{2}}\right)^2 = \frac{1}{\sqrt{2}}\left(\frac{y-x}{\sqrt{2}}\right)$

$\Rightarrow 4a = \frac{1}{\sqrt{2}} \Rightarrow a = \frac{1}{4\sqrt{2}}$

Therefore shortest normal chord = $6.\frac{1}{4\sqrt{2}}\sqrt{3}$

$\Rightarrow p - q = 1$

(S) $y^2 = 4\,(ay^2+2y+1)$

$y^2\,(1-4a) - 8\,y - 4 = 0$

$D = 0$

$64 + 16\,(1-4a) = 0$

$\Rightarrow a = \frac{5}{4}$

SECTION – B

46. 3

Sol. $2 - a = SP$

$= \frac{1}{2}$

$\Rightarrow a = \frac{3}{2} \Rightarrow 2a = 3.$

47. 2

Sol. Points on the first and second lines are (–1, 2, 0) and (3, –4, 1).

Equation of the plane is $8(x + 1) + (y – 2) – 13k(z) = 0$

(3, –4, 1) will also lie on it

$\Rightarrow 8(4) - 6 - 13k = 0$

$\Rightarrow 13k = 26 \Rightarrow k = 2.$

48. 2

Sol. $\frac{dy}{dx} + x\left(\frac{dy}{dx}\right)^2 - y = 0$ ……(1)

Let $y = mx + c$

$m + xm^2 = mx + c$

$\Rightarrow m = c,\ m^2 – m = 0$

$\Rightarrow m = 0, 1$

$y = 0,\ x + 1$

49. 2

Sol. Let $f(x) = x^3 - 3x^2 + 5x = (x-1)^3 + 2(x-1) + 3$

$g(y) = y^3 + 2y \Rightarrow g'(y) = 3y^2 + 2 > 0 \forall y \in R$

$\Rightarrow g(\alpha - 1) = -2$ and $g(\beta - 1) = 2$ and g(y) is odd

$\Rightarrow (\alpha + \beta) = 2$

50. 3

Sol. $f(x) = \sin x + \int_{-\pi/2}^{\pi/2} (\sin x + t\cos x)\left(-\frac{1}{k}\sin t - \frac{2}{k}\cos t\right)dt$

$= \sin x + I_1 + I_2 + I_3 + I_4$

Where $I_1 = -\frac{\sin x}{k}\int_{-\pi/2}^{\pi/2} \sin t\,dt = 0$

$I_2 = -\frac{2\sin x}{k}\int_{-\pi/2}^{\pi/2} \cot dt = -\frac{4\sin x}{k}$

$I_3 = -\frac{\cos x}{k}\int_{-\pi/2}^{\pi/2} t\sin t\,dt = -\frac{2\cos x}{k}$

$I_4 = \frac{4\cos x}{k}\int_{-\pi/2}^{\pi/2} t\cos t\,dt = 0 \qquad (\because t\cos t \text{ is odd})$

$\Rightarrow -\frac{1}{k}\sin x - \frac{2}{k}\cos x = \sin x - \frac{4\sin x}{k} - \frac{2\cos x}{k}$

$\Rightarrow -\frac{1}{k} = 1 - \frac{4}{k} \Rightarrow k = 3.$

51. 1

Sol. $4\int_1^x f(t)dt = 2xf(x) - x^2$

Differentiating both sides f(x) = xf'(x) – x

or $y = x\frac{dy}{dx} - x$

or $\frac{dy}{dx} - \frac{y}{x} = 1$ (Linear form)

On solving $\frac{y}{x} = \ell nx + c$

$\because f(e) = 1 \Rightarrow \frac{1}{e} = 1 + c \Rightarrow c = \frac{1}{e} - 1$

$\frac{y}{x} = \ell nx + \frac{1}{e} - 1$

$y = x\ell nx + \frac{1}{e} - 1$

$y = x\ell nx + \frac{x}{e} - x$

$f(3) = 3\ell n3 + \frac{3}{e} - 3 = 1.394$

$\therefore [f(3)] = 1$